

A Gaussian–Section Obstruction to Uniform Common Depth

A negative range for Problem 2.3 of arXiv:2401.15095

Autonomous proof packet for `fa_banach_001`

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Status. Candidate partial result, likely valid, pending human review. The result gives a broad negative parameter range, not a full classification of all tuples in the source problem.

1 Source problem

Let $n = n_1 + \cdots + n_r$ and let ν_i be the rank $k = n_1 + \cdots + n_i$ tautological bundle over the real flag manifold $\text{Fl}(n_1, \dots, n_r)$. Problem 2.3 of Antolín Camarena and Calles Loperena asks for all tuples for which every collection of m mass assignments on ν_i admits a fiber and a point having “sufficient depth” for all m measures [1, Problem 2.3, PDF p. 4]. The preceding paragraph explains that this means a depth bounded below by a positive constant depending only on the base.

We shall focus on the following problem:

Problem 2.3. *Determine all tuples of positive integers $(m, i, n_1, n_2, \dots, n_r)$ such that for every collection of m mass assignments $\mu_1, \dots, \mu_m$ on ν_i over $\text{Fl}(n_1, \dots, n_r)$, with $1 \leq i \leq r - 1$, there exists a flag*

$$\mathcal{F} = (V_1, V_2, \dots, V_{r-1}) \in \text{Fl}(n_1, n_2, \dots, n_r),$$

with $V_1 \subset V_2 \subset \cdots \subset V_r = \mathbb{R}^n$ and $\dim(V_s) = \sum_{j=1}^s n_j$, and a point $x \in V_i$ with the property that all the measures $\mu_1^{\mathcal{F}}, \dots, \mu_m^{\mathcal{F}}$ on V_i have sufficient depth with respect to the point x .

For a probability measure μ on a Euclidean space, vanishing on affine hyperplanes, recall

$$\text{depth}_{\mu}(x) = \inf\{\mu(H) : H \text{ is a closed halfspace containing } x\}.$$

2 Idea of the proof

A nowhere-zero section of $E^{\oplus(m-1)}$ is a continuous choice of m fiber points, with the first at the origin, that are not all equal. After normalization, two of these centers stay a definite distance apart in every fiber. Put very narrow spherical Gaussians at the centers. Every candidate common point is then far from at least one center, and the Tukey depth of the corresponding Gaussian is uniformly tiny. Thus a topological section becomes an adversarial family of perfectly regular mass assignments.

For the flag tautological bundle, $n - k + 1$ fixed ambient vectors always give such a section by orthogonal projection: they cannot all be perpendicular to a k -plane. A Stiefel–Whitney calculation shows that fewer summands cannot have a nowhere-zero section, making the sectional threshold exact.

3 The general bundle obstruction

Let $E \rightarrow B$ be a rank- k Euclidean vector bundle. Define its best universal simultaneous depth constant for m assignments by

$$D_m(E) = \inf_{(\mu_1, \dots, \mu_m)} \sup_{b \in B, x \in E_b} \min_{1 \leq j \leq m} \text{depth}_{\mu_j^b}(x),$$

where the infimum runs over continuous mass assignments in the sense of the source.

Lemma 1 (spherical Gaussian depth). *Let $\gamma_{c,\sigma}$ be the Gaussian probability measure on $\mathbb{R}^k$ with mean c and covariance $\sigma^2 I$. If Φ is the standard normal distribution function, then*

$$\text{depth}_{\gamma_{c,\sigma}}(x) = \Phi\left(-\frac{\|x - c\|}{\sigma}\right).$$

Proof. For a unit vector u , the least-mass halfspace with inward normal u that contains x has boundary through x . Its mass on the side away from c is

$$\Phi\left(-\frac{\langle x - c, u \rangle}{\sigma}\right).$$

Minimizing over u is the same as maximizing $\langle x - c, u \rangle$, whose maximum is $\|x - c\|$. $\square$

Theorem 1 (Gaussian-section obstruction). *If $E^{\oplus(m-1)}$ has a nowhere-zero continuous section, then*

$$D_m(E) = 0.$$

Equivalently, for every $\eta > 0$ there are m mass assignments such that no fiber point has depth at least η for all of them.

Proof. Write a nowhere-zero section as

$$S(b) = (s_2(b), \dots, s_m(b)) \in E_b^{\oplus(m-1)}$$

and normalize it so that

$$\sum_{j=2}^m \|s_j(b)\|^2 = 1 \quad (b \in B).$$

Set $c_1(b) = 0$ and $c_j(b) = s_j(b)$ for $2 \leq j \leq m$. For each b , some $j \geq 2$ satisfies $\|c_j(b) - c_1(b)\| \geq 1/\sqrt{m-1}$. Consequently, for every $x \in E_b$, the triangle inequality gives

$$\max_{1 \leq a \leq m} \|x - c_a(b)\| \geq \frac{1}{2\sqrt{m-1}}.$$

For $\sigma > 0$, let μ_j^b be the spherical Gaussian on E_b with center $c_j(b)$ and covariance $\sigma^2 I_{E_b}$. In an orthonormal bundle chart, the Gaussian integrals of bounded continuous test functions depend continuously on the center and the fiber frame, so these are continuous mass assignments. They vanish on hyperplanes.

By Lemma 1, for every b and x ,

$$\min_j \text{depth}_{\mu_j^b}(x) \leq \Phi\left(-\frac{1}{2\sigma\sqrt{m-1}}\right).$$

The right side tends to zero as $\sigma \downarrow 0$. Given $\eta > 0$, choose σ making it smaller than η . This proves the claim. $\square$

4 Exact sectional threshold on the flag manifold

Let $B = \text{Fl}(n_1, \dots, n_r)$, let ν_i have fiber V_i , and put

$$n = n_1 + \dots + n_r, \quad k = n_1 + \dots + n_i < n.$$

Proposition 1. *The least positive integer q for which $\nu_i^{\oplus q}$ admits a nowhere-zero section is*

$$q = n - k + 1.$$

Proof. Choose a fixed subspace $W \subset \mathbb{R}^n$ of dimension $n - k + 1$, with orthonormal basis $e_1, \dots, e_{n-k+1}$. At a flag with k -plane V_i , define

$$s_a(V_i) = P_{V_i} e_a \in V_i, \quad 1 \leq a \leq n - k + 1.$$

These sections cannot vanish simultaneously: otherwise $W \subseteq V_i^\perp$, contradicting $\dim W = n - k + 1 > n - k = \dim V_i^\perp$. Hence their tuple is a nowhere-zero section of $\nu_i^{\oplus(n-k+1)}$.

For the converse, the projection

$$\text{Fl}(n_1, \dots, n_r) \longrightarrow G_k(\mathbb{R}^n), \quad (V_1, \dots, V_{r-1}) \longmapsto V_i,$$

pulls the Grassmannian tautological bundle back to ν_i and is injective in mod-2 cohomology, as in the projection used in the source’s proof of its main theorem [1, Section 3]. The standard Grassmannian calculation used there gives

$$w_k(\nu_i)^{n-k} \neq 0.$$

For $1 \leq q \leq n - k$, this implies $w_k(\nu_i)^q \neq 0$. By the Whitney product formula, the top Stiefel–Whitney class of the rank- qk bundle $\nu_i^{\oplus q}$ is

$$w_{qk}(\nu_i^{\oplus q}) = w_k(\nu_i)^q \neq 0.$$

A bundle with a nowhere-zero section has zero top Stiefel–Whitney class. Thus no smaller q works. $\square$

Corollary 1 (negative range for Problem 2.3). *If*

$$m \geq n - k + 2,$$

then for every $\eta > 0$ there are m mass assignments on ν_i such that no flag $\mathcal{F}$ and point $x \in V_i$ give depth at least η for all m measures. Hence no positive “sufficient depth” constant exists in this parameter range.

Proof. Here $m - 1 \geq n - k + 1$. Pad the nowhere-zero section of $\nu_i^{\oplus(n-k+1)}$ from Proposition 1 with zero components to obtain one of $\nu_i^{\oplus(m-1)}$, and apply Theorem 1. $\square$

5 Verification, scope, and novelty

The proof is exact and uses no numerical computation. It gives a negative classification region complementary to the source’s positive characteristic-class theorem. The intermediate range $2 \leq m \leq n - k + 1$ remains open in general; the absence of a nowhere-zero center section there does not itself imply a positive common-depth theorem.

A bounded run-index and arXiv/web search on 17 August 2026 used arXiv id 2401.15095 and combinations of “mass assignments”, “Tukey depth”, “Gaussian centers”, “nowhere-zero section”, and “ $n - k + 2$ ”. It found the source and related positive mass-partition results, but no exact negative statement matching Theorem 1 or Corollary 1. Novelty confidence is moderate.

Human-review recommendation

Review is recommended. The main checks are the exact Gaussian-depth formula, weak continuity of the fiberwise Gaussians, and the mod-2 cohomology injection/nonvanishing used to prove that the sectional threshold is exact. The negative corollary itself only needs the explicit projection section, so it remains valid even if the exact-minimality sentence requires adjustment.

References

- [1] Omar Antolín Camarena and Jaime Calles Loperena, *A Center Transversal Theorem for mass assignments*, arXiv:2401.15095, 2024.